Lecture 16

Connections and geodesics

Objectives. *Differentiating a tangent field involves both its components and its coordinate basis. A connection accounts for this change. A metric selects one connection by compatibility and zero torsion. Its parallel transport preserves inner products, and its geodesics have parallel velocity.*

By the end of this lecture you will be able to:

- Construct the Levi-Civita connection from a nondegenerate metric.
- Compute parallel transport and geodesic equations in coordinates.
- Find a conserved geodesic quantity from a Killing field.
- Explain why the exponential is onto for a compact connected Lie group.

Unit 16A Connections, transport and geodesics

16.1 Affine connections

We first differentiate a constant vector in a moving basis. In the punctured Euclidean plane, the polar unit vectors are

$$e_r = (\cos\phi, \sin\phi), \qquad e_\phi = (-\sin\phi, \cos\phi), \qquad \partial_r = e_r, \quad \partial_\phi = r e_\phi.$$

Along the unit circle $\gamma(t) = (\cos t, \sin t)$, the constant Cartesian vector $V = (1, 0)$ has changing components:

$$V = \cos t\, e_r - \sin t\, e_\phi, \qquad \frac{de_r}{dt} = e_\phi, \quad \frac{de_\phi}{dt} = -e_r.$$

The ordinary product rule includes the derivatives of the basis vectors:

$$\frac{dV}{dt} = -\sin t\, e_r + \cos t\, e_\phi - \cos t\, e_\phi + \sin t\, e_r = 0.$$

Differentiating only the components would miss this cancellation. The figure shows the constant vector in the two polar bases.

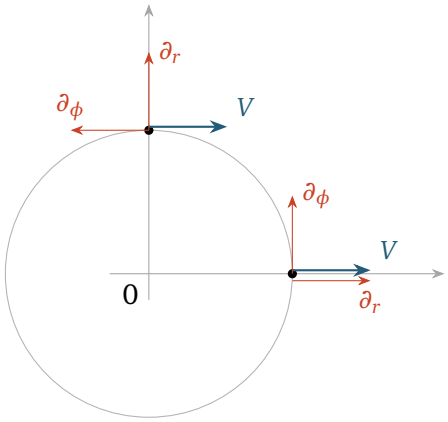


Figure 16.1: The same Cartesian vector $(1, 0)$ at two points of the unit circle. Its polar components change from $(1, 0)$ at $(1, 0)$ to $(0, -1)$ at $(0, 1)$.

On a surface in Euclidean space, a tangent field's ambient derivative can have a normal component. Tangential projection keeps its derivative within the tangent space. On the unit sphere, take the equator $\gamma(t) = (\cos t, \sin t, 0)$ and the field $V(t) = (1 + t)\dot{\gamma}(t)$. At $p = \gamma(0)$ its ambient derivative is $(-1, 1, 0)$, with tangent part $(0, 1, 0)$ and normal part $(-1, 0, 0)$. The figure also shows the field at $q = \gamma(\pi/3)$.

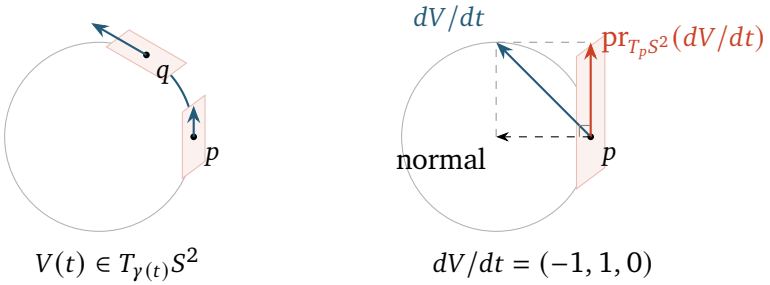


Figure 16.2: Field values at p and q belong to their own tangent planes. At p, the ambient derivative splits into its tangent and normal components.

For the velocity field $V = \dot{\gamma}$ alone, the ambient derivative is $\ddot{\gamma} = -\gamma$. Its tangent projection vanishes. A connection gives a derivative on a manifold without requiring an ambient space.

Let M be a smooth n-manifold without boundary and let $\mathfrak{X}(M)$ denote its smooth vector fields, as in Lecture 9.

Definition 16.1. An *affine connection* is an $\mathbb{R}$-bilinear map

$$\nabla \colon \mathfrak{X}(M) \times \mathfrak{X}(M) \to \mathfrak{X}(M)$$

that satisfies, for $f \in C^\infty(M)$,

$$\nabla_{fX} Y = f \nabla_X Y, \qquad \nabla_X (fY) = X(f) Y + f \nabla_X Y.$$

The field $\nabla_X Y$ is the *covariant derivative* of Y in direction X.

The first rule makes the derivative depend only on the direction at the point. The second rule differentiates scalar multiples of the field by the product rule. For example, on Cartesian $\mathbb{R}^n$, the usual directional derivative $\nabla_X Y = X^i \partial_i Y^j \partial_j$ satisfies both rules.

The rules make the derivative local. Suppose Y vanishes on a neighbourhood of p. Take a cutoff function f at p with support in that neighbourhood (Definition 12.12). Then $fY = 0$, and the Leibniz rule at p gives $\nabla_X Y|_p = 0$. The first rule proves the same locality in X. Thus local fields can be extended with cutoffs before taking their derivative.

In coordinates $x^1, \dots, x^n$, write $\partial_i = \partial/\partial x^i$.

Definition 16.2. The *Christoffel symbols* Γ_{ij}^k are defined by

$$\nabla_{\partial_i}\partial_j = \Gamma_{ij}^k \partial_k, \qquad (\nabla_X Y)^k = X^i(\partial_i Y^k + \Gamma_{ij}^k Y^j).$$

Repeated indices range from 1 to n. The second formula follows from the two connection rules and shows that the symbols determine the connection. In a chart, this expression uses only X_p, Y_p and the first derivatives of Y at p.

Definition 16.3. The *torsion* is

$$T(X,Y) = \nabla_X Y - \nabla_Y X - [X,Y].$$

The connection is *torsion-free* when $T = 0$.

It is a tensor: the derivative term in $\nabla_Y(fX) = Y(f)X + f\nabla_Y X$ cancels the derivative term in $[fX,Y] = f[X,Y] - Y(f)X$. Thus $T(fX,Y) = fT(X,Y)$. Antisymmetry gives linearity over functions in the other slot. To see why function-linearity gives a pointwise tensor, restrict locally using cutoffs. Expanding $Z = Z^i\partial_i$ in any function-linear slot gives $A(Z)(p) = Z^i(p)A(\partial_i)(p)$, which depends only on Z_p.

Since coordinate fields commute, zero torsion is equivalent to $\Gamma_{ij}^k = \Gamma_{ji}^k$.

Definition 16.4. For a pseudo-Riemannian metric g, a *metric connection* satisfies

$$Xg(Y,Z) = g(\nabla_X Y, Z) + g(Y, \nabla_X Z) \qquad (X,Y,Z \in \mathfrak{X}(M)).$$

This identity differentiates the inner product using the connection.

For a parametrized surface $p(x^1,x^2)$, the equality of mixed partials gives $\partial_i\partial_j p = \partial_j\partial_i p$. Tangential projection preserves this equality. Zero torsion expresses this property intrinsically. Metric compatibility alone does not force it. On Euclidean $\mathbb{R}^3$, the connection $\nabla_{\partial_i}\partial_j = \varepsilon_{ijk}\partial_k$ is metric because $\varepsilon_{ijk} + \varepsilon_{ikj} = 0$. Its torsion is $T(\partial_i,\partial_j) = 2\varepsilon_{ijk}\partial_k$. The alternating symbol has $\varepsilon_{123} = 1$.

16.2 The Levi-Civita connection

We derive the connection from metric compatibility and zero torsion. The cyclic sum in Koszul's formula determines it uniquely.

Definition 16.5. A torsion-free metric connection is called the *Levi-Civita connection*.

Theorem 16.6 (Levi-Civita). *Every pseudo-Riemannian manifold has a unique torsion-free metric connection. It satisfies the Koszul formula*

$$\begin{aligned} 2g(\nabla_X Y, Z) = {} & Xg(Y,Z) + Yg(X,Z) - Zg(X,Y) \\ & - g(X,[Y,Z]) + g(Y,[Z,X]) + g(Z,[X,Y]). \end{aligned}$$

Its coordinate symbols are

$$\Gamma_{ij}^k = \frac{1}{2} g^{k\ell}(\partial_i g_{\ell j} + \partial_j g_{\ell i} - \partial_\ell g_{ij}).$$

Proof. For uniqueness, metric compatibility gives

$$\begin{aligned} Xg(Y,Z) &= g(\nabla_X Y, Z) + g(Y, \nabla_X Z), \\ Yg(X,Z) &= g(\nabla_Y X, Z) + g(X, \nabla_Y Z), \\ Zg(X,Y) &= g(\nabla_Z X, Y) + g(X, \nabla_Z Y). \end{aligned}$$

Add the first two identities and subtract the third. Replace $\nabla_Y X$ by $\nabla_X Y - [X,Y]$, and do the same for the other reversed pairs. The result is

$$\begin{aligned} Xg(Y,Z) + Yg(X,Z) - Zg(X,Y) = {} & 2g(\nabla_X Y, Z) - g([X,Y],Z) \\ & + g(Y,[X,Z]) + g(X,[Y,Z]). \end{aligned}$$

Rearranging gives Koszul's formula. Nondegeneracy determines $\nabla_X Y$ uniquely.

For existence, denote the right side by $\mathcal{K}(X,Y,Z)$. Expand derivatives by the product rule and brackets by $[X, fY] = X(f)Y + f[X,Y]$. The derivative terms cancel to give

$$\begin{aligned} \mathcal{K}(X,Y,fZ) &= f\mathcal{K}(X,Y,Z), \\ \mathcal{K}(fX,Y,Z) &= f\mathcal{K}(X,Y,Z), \\ \mathcal{K}(X,fY,Z) &= f\mathcal{K}(X,Y,Z) + 2X(f)g(Y,Z). \end{aligned}$$

For example, in the first identity the extra terms are $X(f)g(Y,Z) + Y(f)g(X,Z) - Y(f)g(X,Z) - X(f)g(Y,Z) = 0$. The cutoff and coordinate-expansion argument of §16.1 therefore makes $\mathcal{K}(X,Y,\cdot)/2$ a smooth covector field. Apply the inverse metric to define $\nabla_X Y$. The remaining identities give linearity in X and the Leibniz rule in Y. Additivity and real bilinearity follow directly from the formula.

Finally, exchanging the indicated slots in $\mathcal{K}$ gives

$$\begin{aligned} \mathcal{K}(X,Y,Z) - \mathcal{K}(Y,X,Z) &= 2g([X,Y],Z), \\ \mathcal{K}(X,Y,Z) + \mathcal{K}(X,Z,Y) &= 2Xg(Y,Z). \end{aligned}$$

The first identity proves zero torsion by nondegeneracy. The second proves metric compatibility. Substituting coordinate fields in Koszul, whose brackets vanish, gives $2g_{k\ell}\Gamma_{ij}^k = \partial_i g_{j\ell} + \partial_j g_{i\ell} - \partial_\ell g_{ij}$. Multiplication by the inverse metric yields the stated symbols. ∎

For a constant metric matrix the symbols vanish in those coordinates. In particular, Minkowski geodesics will be straight lines. For an embedded surface in Euclidean three-space, the connection is $\nabla_X Y = \mathrm{pr}_{TM}(D_X Y)$, the tangential projection of the ambient derivative. Indeed, projection preserves the derivative rules. The ambient product rule gives metric compatibility because normal components pair to zero with tangent fields. The ambient identity $D_X Y - D_Y X = [X,Y]$ gives zero torsion. Uniqueness identifies this connection.

Self-check. Compute the plane's Levi-Civita symbols in polar coordinates $r > 0$. Compare with Cartesian coordinates to test whether Christoffel symbols are a tensor.

Answer. The metric is $g = \mathrm{d}r \otimes \mathrm{d}r + r^2 \mathrm{d}\phi \otimes \mathrm{d}\phi$. Its only nonzero derivative is $\partial_r g_{\phi\phi} = 2r$. The formula gives

$$\Gamma^r_{\phi\phi} = -r, \qquad \Gamma^\phi_{r\phi} = \Gamma^\phi_{\phi r} = 1/r.$$

All other symbols vanish. The Cartesian symbols are zero, whereas these are nonzero at the same points. A tensor with zero components in one basis has zero components in every basis. Thus Christoffel symbols are not tensor components.

16.3 Parallel transport and geodesics

A curve can meet the same point at different parameter values with different field values. Such a field need not extend to a field on a neighbourhood. We define its derivative along the curve and check that the definition agrees in different bases.

Definition 16.7. A *field along a curve* $\gamma\colon [a,b] \to M$ assigns $V(t) \in T_{\gamma(t)}M$ smoothly. Its *covariant derivative along the curve* is given in a coordinate segment by

$$\frac{DV}{dt} = (\dot V^k + \Gamma^k_{ij}(\gamma(t))\dot x^i V^j)\partial_k.$$

Equivalently, write this expression as $\dot v^j \partial_j + v^j \nabla_{\dot\gamma}\partial_j$ for $V = v^j\partial_j$ along the curve. In its first argument, the connection uses only the value of the direction field at the point.

Definition 16.8. A *frame* is a local smooth basis, written as the row $e = (e_1, \dots, e_n)$.

Multiplication of this row by a coefficient column means $ev = \sum_j v^j e_j$. For a new frame $e' = eA$ and coefficient columns $v = Av'$, define C by $\nabla_{\dot\gamma} e = eC$. The Leibniz rule and the coefficient change give

$$\nabla_{\dot\gamma}(eA) = e(\dot A + CA),$$
$$e(A\dot v' + \dot A v' + CAv') = e(\dot v + Cv).$$

The first line differentiates the new basis. The second compares the two expressions for DV/dt. Here $\dot A$ is the derivative along γ. The Leibniz rule supplies this term because $v = Av'$ gives $\dot v = \dot A v' + A\dot v'$. Thus the derivative is independent of the frame, including at repeated points of the curve. For $V = Y \circ \gamma$ it equals $\nabla_{\dot\gamma} Y$.

Definition 16.9. A field is *parallel* when $DV/dt = 0$.

Definition 16.10. The *parallel transport* along $\gamma\colon [a,b] \to M$ sends an initial vector to the endpoint of its parallel extension, written

$$P_\gamma\colon T_{\gamma(a)}M \to T_{\gamma(b)}M, \qquad P_\gamma(v) = V(b), \quad V(a) = v, \quad DV/dt = 0.$$

Proposition 16.11 (Parallel transport · *proved given linear ODE existence and uniqueness*). *For a smooth curve $\gamma\colon [a,b] \to M$ and $v \in T_{\gamma(a)}M$, there is a unique parallel field with $V(a) = v$. The map P_γ is a linear isomorphism. For a metric connection it preserves the metric.*

Proof. The coordinate equation is a linear ODE $\dot{V} = -C(t)V$, where $C^k{}_j = \Gamma^k_{ij}\dot{x}^i$. On a compact coordinate segment its smooth coefficients are bounded, so linear ODE existence and uniqueness give a solution throughout that segment. A finite subdivision places each closed segment in a chart. Propagate the initial value through the subdivision. Coordinate independence and uniqueness glue the solutions. The same uniqueness proves linearity. Reversing the curve gives the inverse map. In coordinates, metric compatibility reads $\partial_i g_{k\ell} = \Gamma^m_{ik} g_{m\ell} + \Gamma^m_{i\ell} g_{km}$. Differentiate $g_{k\ell}V^kW^\ell$ along the curve and substitute this identity. It gives the along-curve product rule. For parallel fields,

$$\frac{d}{dt} g(V,W) = g(DV/dt, W) + g(V, DW/dt) = 0.$$

Thus endpoint inner products agree. ∎

This is the ODE input already used in Theorem 9.8.

Example 16.12 (Transport around a plane circle)**.** We transport the initial vector with polar components $(1,0)$ along $r = 1$ and $\phi = t$. The polar symbols computed above give

$$\dot{V}^r - V^\phi = 0, \qquad \dot{V}^\phi + V^r = 0.$$

The solution with these initial data is

$$V^r(t) = \cos t, \qquad V^\phi(t) = -\sin t.$$

Substitution gives $-\sin t - (-\sin t) = 0$ and $-\cos t + \cos t = 0$. Reconstruction gives $V = \cos t\, e_r - \sin t\, e_\phi = (1,0)$. Thus the changing polar components describe a constant Cartesian vector.

Definition 16.13. A *geodesic* is a parametrized curve with $D\dot{\gamma}/dt = 0$.

A geodesic has zero covariant acceleration. On an embedded surface, its ambient acceleration is normal to the surface. Its coordinate equation is

$$\ddot{x}^k + \Gamma^k_{ij}\dot{x}^i\dot{x}^j = 0.$$

The local second-order ODE theorem gives a unique maximal geodesic γ_v with $\gamma_v(0) = p$ and $\dot{\gamma}_v(0) = v \in T_pM$. Here *maximal* means that its interval of definition admits no extension. The geodesic equation is a first-order field on TM, with bundle components $(\dot{x}^k, \dot{v}^k) = (v^k, -\Gamma^k_{ij}v^iv^j)$.

Definition 16.14. The *connection exponential* is the map

$$\operatorname{Exp}_p \colon \mathcal{D}_p \to M, \qquad \operatorname{Exp}_p(v) = \gamma_v(1),$$

where $\mathcal{D}_p \subset T_pM$ consists of initial velocities for which the geodesic exists through time 1.

Definition 16.15. The connection is *geodesically complete* if every maximal geodesic is defined on all of $\mathbb{R}$.

The open smooth flow domain in Theorem 9.8 makes this map smooth. The domain is open as the time-1 slice of that flow domain, restricted to T_pM. It contains zero because the constant curve is a geodesic. For a geodesically complete connection, $\mathcal{D}_p = T_pM$. For the Levi-Civita connection, $g(\dot{\gamma}, \dot{\gamma})$ is constant by metric compatibility. In the Riemannian case, its square root is the speed.

Definition 16.16. The *energy* of a curve on $[a, b]$ is

$$E(\gamma) = \frac{1}{2}\int_a^b g(\dot\gamma, \dot\gamma)\, dt.$$

A curve is *stationary* when $\frac{d}{ds}\big|_0 E(\gamma_s) = 0$ for every smooth family γ_s with $\gamma_0 = \gamma$ and fixed endpoints.

Physical input. For a free unit-mass particle constrained to a Riemannian manifold, with no potential, we take the kinetic Lagrangian to be $L = \frac{1}{2}g_{ij}\dot x^i \dot x^j$. The physical paths satisfy the Euler–Lagrange equations recalled from *Theoretical Mechanics*, Lecture 2, *Lagrange's equations*.

Proposition 16.17 (Energy equations · *sketched; coordinate equivalence proved*)**.** *The Levi-Civita geodesics are precisely the stationary curves of the energy with fixed endpoints.*

The Euler–Lagrange equation is $d(\partial L/\partial \dot x^k)/dt - \partial L/\partial x^k = 0$. For $L = \frac{1}{2}g_{ij}\dot x^i \dot x^j$, its left side is

$$g_{k\ell}\ddot x^\ell + \partial_i g_{kj}\dot x^i \dot x^j - \frac{1}{2}\partial_k g_{ij}\dot x^i \dot x^j = g_{k\ell}(\ddot x^\ell + \Gamma^\ell_{ij}\dot x^i \dot x^j).$$

The contraction uses the symmetry of $\dot x^i \dot x^j$:

$$\partial_i g_{kj}\dot x^i \dot x^j = \frac{1}{2}(\partial_i g_{kj} + \partial_j g_{ki})\dot x^i \dot x^j.$$

Nondegeneracy gives the geodesic equation. For a fixed-endpoint variation, write $W^k = \partial_s x^k_s\big|_{s=0}$. Differentiating the energy and integrating by parts gives

$$\begin{aligned}\left.\frac{dE(\gamma_s)}{ds}\right|_0 &= \int_a^b \left(\frac{1}{2}\partial_k g_{ij} W^k \dot x^i \dot x^j + g_{kj}\dot W^k \dot x^j\right) dt \\ &= [g_{kj}W^k \dot x^j]_a^b - \int_a^b g_{k\ell}(\ddot x^\ell + \Gamma^\ell_{ij}\dot x^i \dot x^j) W^k\, dt.\end{aligned}$$

The boundary term vanishes since $W(a) = W(b) = 0$. Arbitrary compactly supported coordinate variations make the integral vanish exactly when its coefficient vanishes. Subdivision into coordinate segments gives the same conclusion for the whole curve. Hence the free particle of the physical input moves along Levi-Civita geodesics, with constant speed.

Definition 16.18. On a connected Riemannian manifold, use Lecture 15's length $\ell(\gamma)$ of a piecewise smooth curve. The distance between two points is the infimum of lengths of piecewise smooth curves joining those points. A *minimizing geodesic* attains that infimum.

For $b > a$, apply Cauchy–Schwarz to the constant function 1 and the speed. This gives

$$\ell(\gamma)^2 \le 2(b-a)E(\gamma),$$

with equality exactly at constant speed. Energy therefore also selects a parametrization. Stationary energy does not imply a global length minimum. No minimizing assertion follows in indefinite signature.

On the unit sphere, tangential projection gives $D\dot\gamma/dt = \ddot\gamma + |\dot\gamma|^2\gamma$. Indeed, differentiating $\gamma \cdot \gamma = 1$ twice gives $\gamma \cdot \ddot\gamma = -|\dot\gamma|^2$. For $v \in T_pS^2$ with $c = |v| > 0$, the unique solution is

$$\gamma_v(t) = \cos(ct)p + \frac{\sin(ct)}{c}v.$$

It lies on the sphere, has the prescribed initial data, and satisfies $\ddot{\gamma} = -c^2\gamma$. Its image is a *great circle*, the intersection with a plane through the origin. For $v = 0$ the geodesic is constant. Evaluating at time 1 gives the sphere exponential

$$\operatorname{Exp}_p(v) = \cos|v|\, p + \frac{\sin|v|}{|v|} v \quad (v \neq 0), \qquad \operatorname{Exp}_p(0) = p.$$

The figure follows radial lines of initial velocities into meridians.

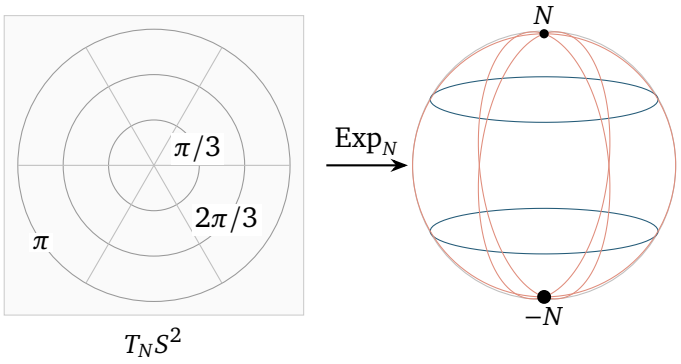


Figure 16.3: At the north pole, tangent circles of radii $\pi/3$ and $2\pi/3$ map to latitudes. The whole circle of radius π maps to the south pole.

In Minkowski coordinates, the zero Christoffel symbols instead give $\gamma_v(t) = p + tv$ for every initial vector, including zero and *null vectors*, which satisfy $\eta(v, v) = 0$.

16.4 Killing fields and conserved quantities

A metric symmetry gives a conserved pairing with geodesic velocity. For a field K with local flow φ_t, define the metric Lie derivative by $L_K g = \frac{d}{dt}\big|_0 \varphi_t^* g$.

Definition 16.19. A *Killing field* satisfies $L_K g = 0$.

A useful coordinate test is independence of a metric coordinate. If $\partial_a g_{ij} = 0$ for all i and j, translation in x^a preserves the metric, so $K = \partial_a$ is Killing. For the kinetic Lagrangian its pairing is the corresponding momentum:

$$g(\partial_a, \dot{\gamma}) = g_{aj}\dot{x}^j = \frac{\partial L}{\partial \dot{x}^a}.$$

A coordinate absent from L is called *cyclic*. The flow law gives $\frac{d}{dt}\varphi_t^* g = \varphi_t^*(L_K g)$, as for forms in Lecture 11. Thus the local flow of a Killing field consists of isometries, and conversely. For local fields Y and Z, put $Y_t = (\varphi_{-t})_* Y$ and $Z_t = (\varphi_{-t})_* Z$. The definition of tensor pullback gives $g(Y, Z) \circ \varphi_t = (\varphi_t^* g)(Y_t, Z_t)$. Differentiate at zero. The product rule and Theorem 9.12 give $Kg(Y, Z) = (L_K g)(Y, Z) + g([K, Y], Z) + g(Y, [K, Z])$. Rearranging proves $(L_K g)(Y, Z) = Kg(Y, Z) - g([K, Y], Z) - g(Y, [K, Z])$. Metric compatibility and zero torsion turn this into

$$(L_K g)(Y, Z) = g(\nabla_Y K, Z) + g(Y, \nabla_Z K).$$

Hence K is Killing exactly when the bilinear form $(Y, Z) \mapsto g(\nabla_Y K, Z)$ is antisymmetric.

Proposition 16.20 (Killing conservation). *Along a Levi-Civita geodesic γ, every Killing field K gives the constant quantity $g(K, \dot\gamma)$.*

Proof. Metric compatibility and the geodesic equation give

$$\frac{d}{dt} g(K, \dot\gamma) = g(\nabla_{\dot\gamma} K, \dot\gamma) = \frac{1}{2}(L_K g)(\dot\gamma, \dot\gamma) = 0. \qquad \blacksquare$$

For example, a constant translation field in Minkowski space is Killing because the metric is constant. Its pairing with $\dot\gamma = v$ is constant, as the straight-line solution also shows.

16.5 The round sphere in coordinates

For the unit sphere, the spherical parametrization is

$$p(\theta, \phi) = (\sin\theta\cos\phi, \sin\theta\sin\phi, \cos\theta), \qquad 0 < \theta < \pi.$$

Here ϕ lies in an angular chart. Lecture 15 computed its induced metric:

$$(g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & \sin^2\theta \end{pmatrix}.$$

The figure compares a great circle with a latitude.

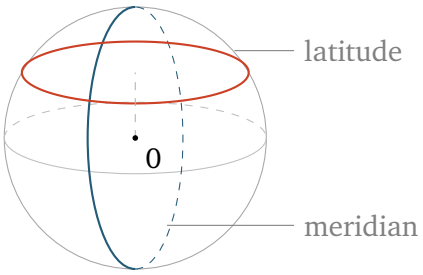


Figure 16.4: A meridian lies in a plane through the sphere's centre. A northern latitude lies in a plane parallel to the equator, away from the centre.

Example 16.21 (Sphere connection and angular momentum). Only $\partial_\theta g_{\phi\phi} = 2\sin\theta\cos\theta$ is nonzero among the metric derivatives. The Christoffel formula gives

$$\Gamma^\theta_{\phi\phi} = -\frac{1}{2}\partial_\theta g_{\phi\phi} = -\sin\theta\cos\theta,$$
$$\Gamma^\phi_{\theta\phi} = \Gamma^\phi_{\phi\theta} = \frac{1}{2\sin^2\theta}\partial_\theta g_{\phi\phi} = \cot\theta.$$

All other symbols vanish. Substitution gives

$$\ddot\theta - \sin\theta\cos\theta\,\dot\phi^2 = 0, \qquad \ddot\phi + 2\cot\theta\,\dot\theta\dot\phi = 0.$$

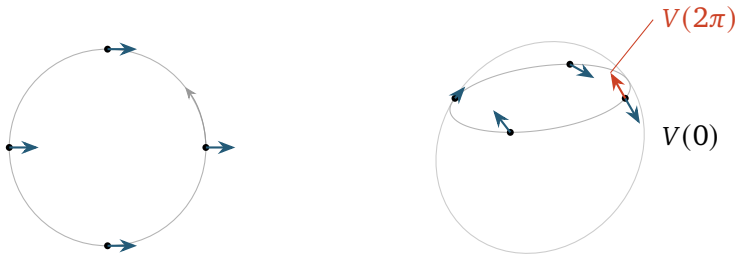
For the latitude circle $\theta = \theta_0$ traversed at constant nonzero angular speed, the first equation requires $\sin\theta_0\cos\theta_0 = 0$. Within this chart only $\theta_0 = \pi/2$ works: the equator is a geodesic. The coordinate poles are points, not nonconstant latitude curves. A meridian with constant ϕ and constant $\dot\theta$ solves both equations: $\ddot\theta = 0$ and $\dot\phi = \ddot\phi = 0$.

The field $K = \partial_\phi$ generates rotations about the vertical axis, which preserve the metric. Its conserved quantity and its derivative are

$$g(K, \dot\gamma) = \sin^2\theta\,\dot\phi, \qquad \frac{d}{dt}(\sin^2\theta\,\dot\phi) = \sin^2\theta(\ddot\phi + 2\cot\theta\,\dot\theta\dot\phi) = 0.$$

This is the axial angular momentum for the unit-mass energy Lagrangian. The coordinate singularity at a pole does not make the smooth connection singular.

For a latitude we use the positively oriented orthonormal frame $e_\theta = \partial_\theta$ and $e_\phi = (\sin\theta)^{-1}\partial_\phi$. The following figure compares transport around a plane circle and a spherical latitude.



plane: $V = (1, 0)$ sphere: $a = \pi/3$

Figure 16.5: The plane arrows are constant Cartesian vectors. On the latitude $a = \pi/3$, the solution's frame angle is $-t/2$, and the returning vector is reversed.

Example 16.22 (Transport around a latitude)**.** We transport $V(0) = e_\theta$ along $\theta = a$ and $\phi = t$, with $0 < a < \pi$. The spherical Christoffel symbols give

$$\dot V^\theta - \sin a\cos a\,V^\phi = 0, \qquad \dot V^\phi + \cot a\,V^\theta = 0.$$

Put $W = \sin a\,V^\phi$, the component along e_ϕ. The equations and their solution become

$$\dot V^\theta = \cos a\,W, \qquad \dot W = -\cos a\,V^\theta, \qquad (V^\theta, W) = (\cos(t\cos a), -\sin(t\cos a)).$$

Both differential equations follow by differentiating this pair. The frame returns to its initial value after one circuit. Thus the vector returns rotated by $-2\pi\cos a$, modulo 2π, relative to the positive frame (e_θ, e_ϕ). At the equator the frame components stay fixed. At $a = \pi/3$ the angle is $-\pi$.

The sphere's rotational symmetries also give the conserved vector $\gamma \times \dot\gamma$. We recall the rotation fields from Lecture 9. For the field $K_b(p) = b \times p$, with fixed $b \in \mathbb{R}^3$, its pairing is $g(K_b, \dot\gamma) = b \cdot (\gamma \times \dot\gamma)$. The figure shows the resulting bound on the latitude of a tilted great circle.

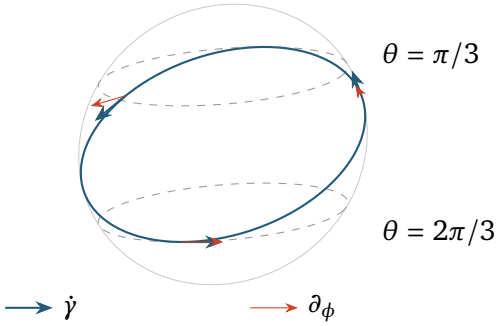


Figure 16.6: The unit-speed great circle $\gamma(t) = (\cos t, \frac{\sqrt{3}}{2}\sin t, \frac{1}{2}\sin t)$ has $\sin^2\theta\,\dot\phi = \sqrt{3}/2$ and $\pi/3 \le \theta \le 2\pi/3$.

For this curve $z = \frac{1}{2}\sin t$, so the polar angle has the stated range. The axial pairing is $x\dot y - y\dot x = \sqrt{3}/2$.

A geodesic need not minimize length on its entire interval. The following great-circle arcs have the same endpoints.

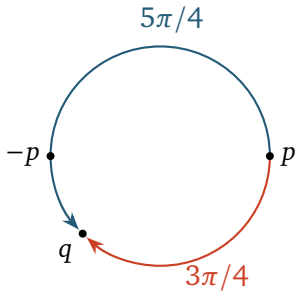


Figure 16.7: From p to q, the increasing-angle unit-circle arc has length $5\pi/4$. The decreasing-angle arc has length $3\pi/4$. The longer geodesic has passed the antipode $-p$.

The antipode is reached after length π along a unit-speed great circle. For an arc of length between π and 2π, the complementary arc has length $2\pi - \ell < \ell$. The longer arc in the figure therefore cannot minimize length. No local geodesic equation implies a global minimum.

Self-check. On the unit sphere, transport $V(0) = 2e_\theta + e_\phi$ along $\theta = \pi/3$ from $\phi = 0$ to $\phi = 2\pi/3$. Give its final coordinate components and check its norm.

Answer. The orthonormal components rotate through $-(2\pi/3)\cos(\pi/3) = -\pi/3$. Thus

$$(V^\theta, W) = \left(1 + \frac{\sqrt{3}}{2}, \frac{1}{2} - \sqrt{3}\right), \qquad V^\phi = \frac{W}{\sin(\pi/3)} = \frac{1}{\sqrt{3}} - 2.$$

The squared norm is $(V^\theta)^2 + W^2 = 5$, equal to $2^2 + 1^2$ initially.

Unit 16B Geodesics and exponentials on Lie groups

16.6 Bi-invariant metrics and the group exponential

For a compact connected Lie group, every element is the exponential of an initial tangent vector at the identity. We prove this by identifying group exponentials with geodesics for a bi-invariant metric. Averaging constructs such a metric, and the cited completeness theorem joins the identity to every point.

On a Lie group G, write $\ell_h(g) = hg$ and $r_h(g) = gh$.

Definition 16.23. A metric is *bi-invariant* if every left and right translation is an isometry.

Recall from Lecture 9 that each $X \in \mathfrak{g} = T_eG$ defines a complete left-invariant field X^L, whose identity curve $c_X(t)$ is a one-parameter subgroup.

Definition 16.24. The *group exponential* is the map

$$\exp\colon \mathfrak{g} \to G, \qquad \exp X = c_X(1).$$

Scaling the ODE gives $c_X(t) = \exp(tX)$, and the flow of X^L is $r_{\exp(tX)}$. This map differs in definition from Exp_p. For a matrix Lie group it agrees with the matrix exponential of Lecture 9.

Theorem 16.25 (Bi-invariant geodesics · *proved given smooth ODE uniqueness*)**.** *For a bi-invariant pseudo-Riemannian metric, $\nabla_{X^L} Y^L = \frac{1}{2}[X,Y]^L$. The geodesics through the identity are exactly $t \mapsto \exp(tX)$, and $\operatorname{Exp}_e = \exp$ on all of $\mathfrak{g}$.*

Proof. The inner products of left-invariant fields are constant. Right translations are isometries, so X^L is Killing. The metric Lie derivative identity above therefore gives $g([X^L,Y^L],Z^L) + g(Y^L,[X^L,Z^L]) = 0$. Using this identity in the three bracket terms of Koszul gives

$$2g(\nabla_{X^L}Y^L, Z^L) = g([X^L,Y^L],Z^L).$$

The fields Z^L span each tangent space, so nondegeneracy proves the formula. Along c_X, its velocity is X^L, hence its acceleration is $\frac{1}{2}[X,X]^L = 0$. Uniqueness of the geodesic ODE gives all identity geodesics. Completeness of X^L gives their full real domains and the exponential equality. ∎

For compact G, a bi-invariant Riemannian metric exists. We construct it using Lecture 12's integration. A left-translated basis of $\mathfrak{g}$ gives a global frame and a positive, left-invariant volume form μ, normalized to have integral 1. Let b be any positive inner product on $\mathfrak{g}$. Define Ad_h to be the differential at the identity of $g \mapsto hgh^{-1}$. Average by

$$B(X,Y) = \int_G b(\operatorname{Ad}_{g^{-1}} X, \operatorname{Ad}_{g^{-1}} Y)\, \mu(g).$$

This is positive definite: a nonzero vector has a strictly positive integrand. Conjugation composes, so substituting $g = hk$ and using left-invariance of volume gives $B(\operatorname{Ad}_h X, \operatorname{Ad}_h Y) = B(X,Y)$. Extend B by left translation to a smooth metric. The identity $r_h \circ \ell_g = \ell_{gh} \circ (g' \mapsto h^{-1}g'h)$ shows that right translation acts on left-translated components by $\operatorname{Ad}_{h^{-1}}$. The Ad-invariance of B proves that the metric is bi-invariant.

For a connected Riemannian manifold, the distance defined in Unit 16A is the infimum of lengths of joining curves. The cited Hopf–Rinow theorem states that completeness of this distance is equivalent to geodesic completeness, and that on a complete connected Riemannian manifold every two points are joined by a minimizing geodesic (Tu, Theorem 16.6, p. 131). The distance induces the manifold topology (Lee, Lemma 6.2, pp. 94–95), so compactness makes it complete. Reparametrize the joining geodesic to $[0,1]$ by a linear change of time. Thus a compact connected G has every element $g = \operatorname{Exp}_e v = \exp v$ for some $v \in \mathfrak{g}$. This surjectivity conclusion is proved given integration and Hopf–Rinow; it does not assert injectivity of the exponential.

Example 16.26 (The rotation group $SO(2)$)**.** The angular metric on $SO(2)$ is invariant under left and right translations, which add a constant angle. For the generator

$$X = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad \exp(tX) = \begin{pmatrix} \cos t & -\sin t \\ \sin t & \cos t \end{pmatrix}.$$

Every plane rotation occurs in this formula. The parameters t and $t + 2\pi$ give the same element, so the exponential is onto and is not injective. The identity geodesics are exactly these constant-angular-speed rotations.

Self-check. State in one sentence what a connection does and why we introduced it. Explain how the two exponential maps become equal at a Lie group's identity.

Answer. A connection differentiates tangent fields by including the change of basis, so that differentiation is independent of the chosen coordinates. For a bi-invariant metric, $\nabla_{X^L} X^L = \frac{1}{2}[X,X]^L = 0$. The group curve $\exp(tX)$ is therefore the geodesic with initial velocity X. Uniqueness gives $\operatorname{Exp}_e X = \exp X$ at time 1.

Summary. A nondegenerate metric determines the torsion-free metric connection by Koszul's formula. Parallel transport solves a linear ODE and preserves the metric. Geodesics have parallel velocity and constant metric norm. Killing fields give conserved pairings with that velocity. For a bi-invariant metric the identity geodesics are one-parameter subgroups. Compact averaging and the cited Hopf–Rinow theorem make the group exponential onto for compact connected groups. Lecture 17 computes curvature from this connection.

Reading. Tu, *Differential Geometry*, §6, §13–§15, especially Problems 15.4–15.5, and Theorem 16.6, p. 131. Lee, *Riemannian Manifolds: An Introduction to Curvature*, Theorem 5.4, pp. 68–70, and Lemma 6.2, pp. 94–95. Nakahara, §7.2, §7.4 and §7.7.